

Periodontology 1

Anatomy and Pathogenesis

ODON 621

Jennifer Goss, DDS
Assistant Professor



Mission: To advance the science and art of health through
education, service, scholarship and social accountability

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Learning Objectives

- 1- Describe macroscopic and microscopic anatomy of periodontal tissues (tooth, attachment, bone)
- 2-Apply knowledge to visible anatomy during a clinical exam
- 3-Recognize changes that occur during disease process
- 4-Understand inflammatory response to bacteria
- 5-Understand immune response to bacteria
- 6-Understand the role of host response in pathogenesis

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In this session, we cover the following topics:

Classification at a Glance

Periodontal Health, Gingival Diseases and Conditions

- Periodontal Health and Gingival Health
- Gingivitis: Dental Biofilm-Induced
- Gingival Diseases: Non-Dental Biofilm-Induced

Periodontitis

- Necrotizing Periodontal Diseases
- Periodontitis
- Periodontitis as a Manifestation of Systemic Disease

Periodontal Manifestations of Systemic Diseases and Developmental and Acquired Conditions

- Systemic Diseases or Conditions Affecting Periodontal Supporting Tissues
- Mucogingival Deformities and Conditions
- Traumatic Occlusal Forces
- Tooth- and Prosthesis-Related Factors
- Periodontal Abscesses and Endodontic-Periodontal Lesions

Peri-Implant Diseases and Conditions

- Peri-Implant Health
- Peri-Implant Mucositis
- Peri-Implantitis
- Peri-Implant Soft and Hard Tissue Deficiencies

2018 Classification of Periodontal and
Peri-Implant Diseases and Conditions



When we discussing inflammation, infection, attachment loss and pathogenesis-I'm going to put them in the framework of the diseases we will be most commonly treating (gingivitis and periodontitis)

Health vs. Disease



How do we get from health to disease? Look at normal so we can identify disease. Last week in our cases we looked at examples of severely inflamed tissues. What happens during inflammation? Tissue destruction?=pathogenesis, and how we can change or reverse that process with therapy

Variations in Anatomy



Can affect the pathogenesis, diagnosis, treatment-things such as phenotype-thick tissue responds differently than thin tissue (bone and soft tissue)

Changes in Tissue: Anatomy

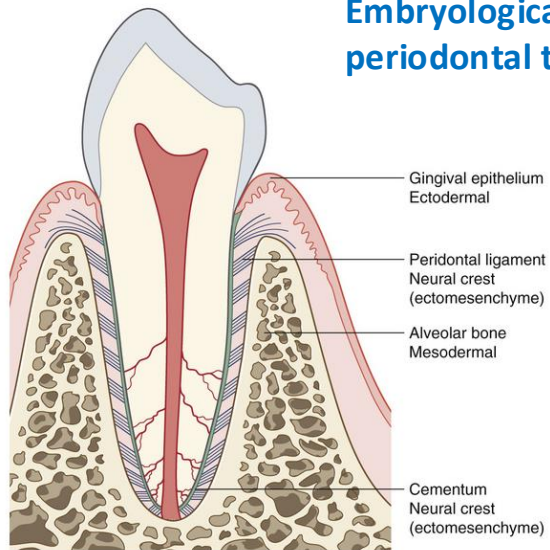


Gingival Anatomy

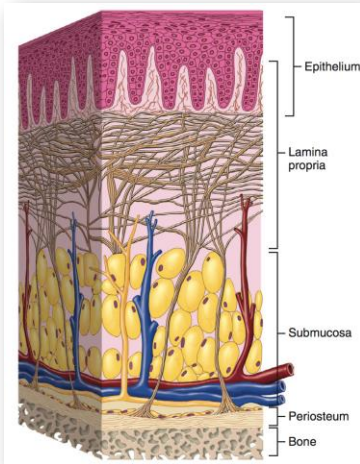
- Outer Gingival Epithelium
- Gingival connective tissue (CT)
- Fibers (dentogingival, circular, transeptal)
- **Gingival margin**
- Gingival groove
- Sulcular epithelium
- Free gingiva
- **Attached gingiva**
- **Mucogingival junction**
- Junctional epithelium

The 3 anatomical features in red can be seen in a visual assessment and used as clinical information

Embryological origins of periodontal tissues



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The Oral Mucosa

= Stratified Squamous Epithelium

1. Lining mucosa: lips, cheeks, floor of the mouth, soft palate, ventral tongue

(non-keratinized)

2. Masticatory mucosa: gingiva, alveolar ridges, hard palate

(keratinized)

(3. Specialized: dorsum of the tongue)

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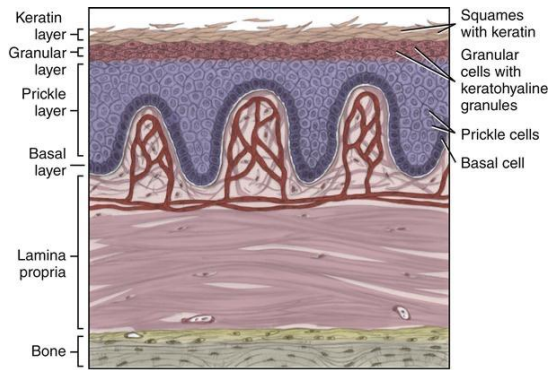
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The oral cavity is lined with stratified squamous epithelium, which is divided into 3 types of tissue.

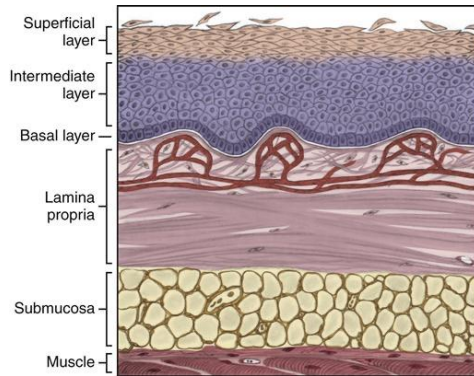
1. Lining mucosa is non-keratinized and covers the floor of the mouth and the cheeks, lips, and soft palate.
2. Masticatory mucosa is keratinized and covers the hard palate, gingiva and alveolar ridges. It's termed masticatory because it comes in primary contact with food during chewing.
3. Specialized mucosa, which covers the dorsal surface of the tongue is keratinized and functions in mastication and taste.

The Oral Epithelium



Orthokeratinized Stratified Squamous Epithelium
(and deeper tissues)

Masticatory Mucosa



Nonkeratinized Stratified Squamous Epithelium
(and deeper tissues)

Lining Mucosa

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To quickly go over differences between masticatory and lining mucosa, you can compare these 2 photos.

The interface between the epithelium and lamina propria in masticatory mucosa is highly interdigitated with more of these rete ridges and connective tissue papillae, giving it a firm base.

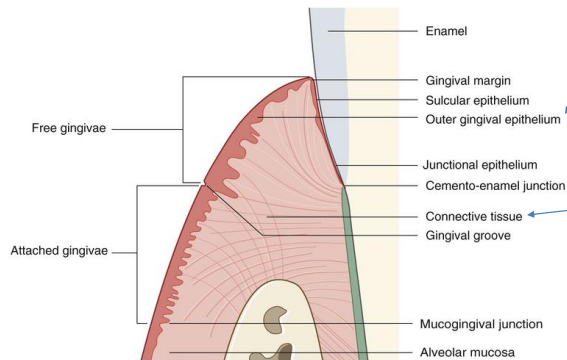
They both have a basal layer.

In lining mucosa, the spinosum or prickly layer is called the intermediate layer.

The granular layer isn't present in lining mucosa.

The keratin layer of the masticatory mucosa is paralleled by the superficial layer in lining mucosa.

Gingival Anatomy



1. Outer gingival epithelium:

Orthokeratinized stratified squamous epithelium

Keratinization= resistant to abrasion and food, inflammation

Covers fibrous CT =supracrestal attachment ("Biologic Width Att" is JE + CT)

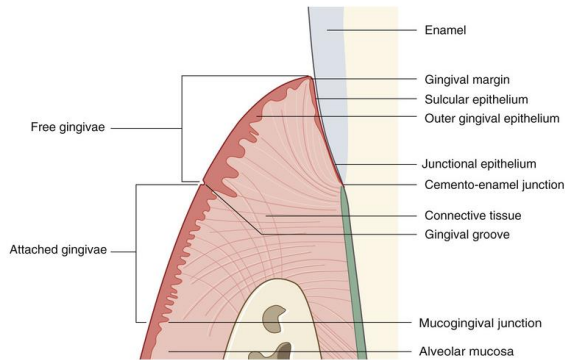
2. (Gingival) Connective tissue:

mesh collagen fibers in ECM (extracellular matrix)

Fibroblasts, blood vessels, immune system cells

Mostly Type I collagen (also III, IV)

Gingival Anatomy



Free gingiva (FG)

cuff of keratinized gingiva above alveolar crest,

interproximally forms the papilla and col

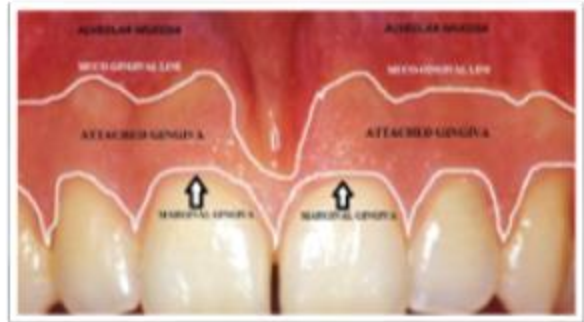
Attached gingiva (AG)

keratinized gingiva that is bound to alveolar bone

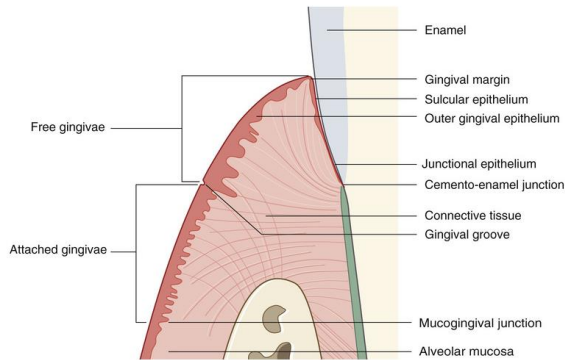
*Mucogingival junction (MGJ)

line that marks boundary between AG and mucosa (darker red: more blood vessels, non-keratinized)

Gingival anatomy-clinical



Gingival Anatomy



- Sulcular epithelium
 - Lines inside of sulcus, para-keratinized, rapid cell turnover
 - Can be non-keratinized ***
- Junctional epithelium
 - Contact gingiva and tooth
 - Attaches by hemidesmosomes
 - Fragile, no rete ridges
 - Permeable

***changed slide-can be para or non keratinized, DOES not have rete ridges

Blood supply and Gingival crevicular fluid (GCF)

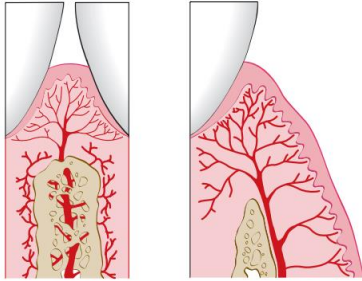
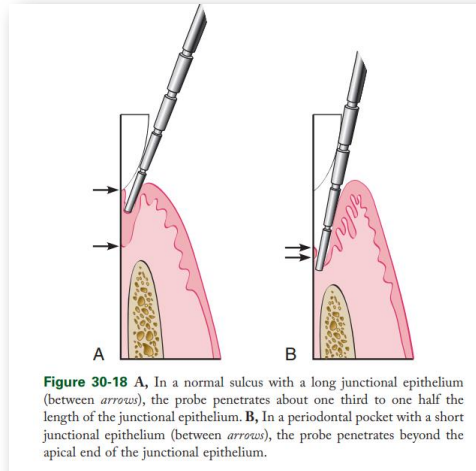


Figure 2-22 Diagram of arteriole penetrating the interdental alveolar bone to supply the interdental tissues (*left*) and a suprapariosteal arteriole overlying the facial alveolar bone, sending branches to the surrounding tissue (*right*).

- GCF contains cells, inflammatory cells, serum, bacteria
- Transudate or exudate
- Increases with inflammation
- Functions: clear material from sulcus, possess antimicrobial properties
- Diagnostic/prognostic value?
- Changes during inflammatory process

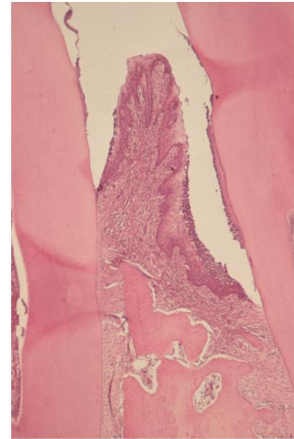
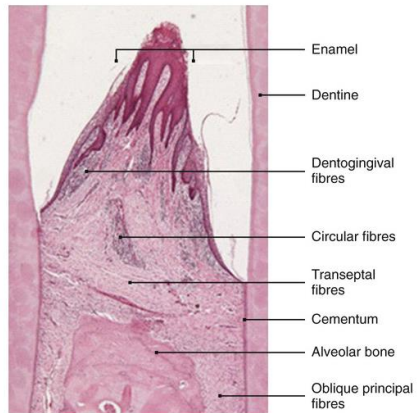
Histology: Correlation to clinical findings



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(From Carranza text book)

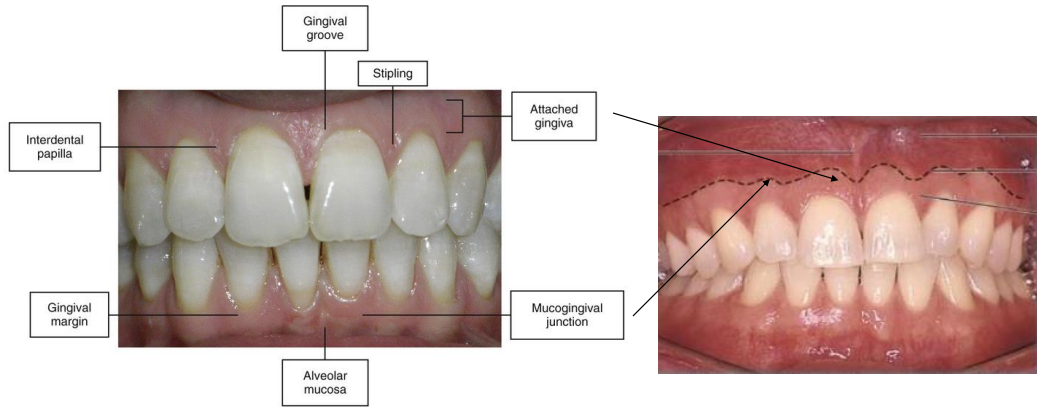


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Image on left is healthy, image on right is a periodontal "pocket" , see dense inflammatory cell infiltrate, bone loss

Anatomy



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Be able to correlate the microanatomy with what we see clinically it will help with....
(next slide)

Identify and describe: Mucogingival Deformity



What do you see?

Identify: CEJ

Color (root vs crown)

Line between enamel & cementum

Proximal teeth: length/position

Identify:

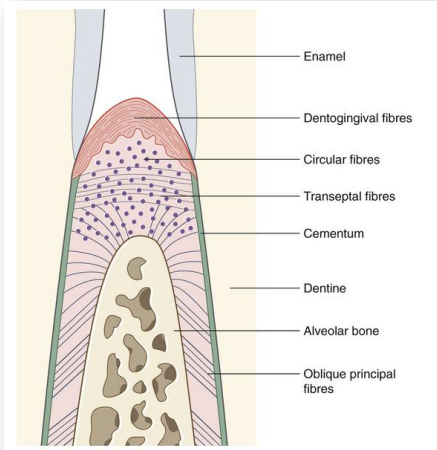
Gingival margin (GM)

Mucogingival junction (MGJ)

Width of keratinized gingiva (KG)

Be able to identify normal anatomical markers (the CEJ, Gingival Margin, Mucogingival Junction, Attached Gingiva. Exposed cementum is at risk for hypersensitivity, decay, abrasion and NCCL (non carious cervical lesions)
IS the pathogenesis for this condition the same as gingivitis, periodontitis?

Anatomy of PDL, Cementum and Alveolar Bone



- Gingival Fibers (dentogingival, circular, transeptal)
 - physical barrier to bacteria, inflammation
 - Not present on dental implants
- Oblique fibers part of PDL

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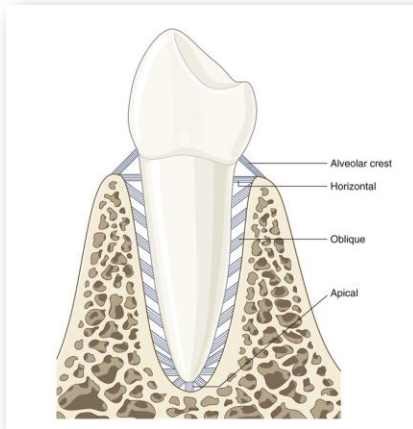
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PDL=Periodontal ligament

Supracrestal Attachment

- *Supracrestal attached tissues* are composed of the **junctional epithelium and the supracrestal connective tissue attachment**. This was formally referred to as the *biologic width*.
- The apico-coronal dimension of the supracrestal attached tissues is variable.
- Clinically, there is evidence that placement of restorative margins within the supracrestal connective tissues is associated with inflammation and loss of periodontal supporting tissues

Periodontal Ligament (PDL)



- Fibrous attachment between tooth and bone, collagen fibers insert into cementum and bone
- Maintains tooth position
- Resists occlusal loading
- Maintains and repairs alveolar bone & cementum (stem cells)
- Neurological functions of mastication through mechanoreceptors

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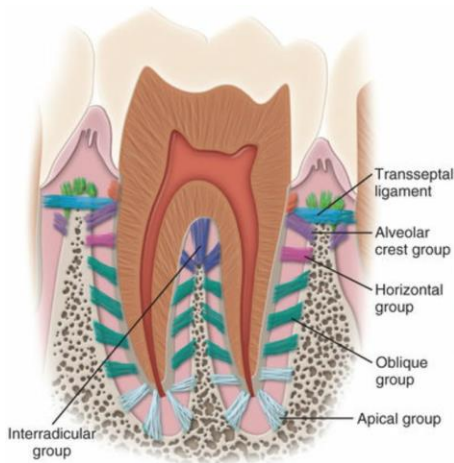
Very important functions!! Lot of implications for function, repair, chewing, and regeneration

Organization of the PDL

Principal Fibers

Collagen I, III, XII

1. Apical
2. Oblique
3. Horizontal
4. Alveolar Crest
5. Interradicular
6. Transseptal



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The predominant collagens of the PDL are types I, III, and XII.

These collagen fibrils are arranged in definite and distinct fiber bundles...the bundles running between tooth and bone, the first 5 listed here, are the called “dentoalveolar fibers”

Resident Cells of the PDL

Fibroblasts – principle cells of the PDL, most numerous and surrounded by intercellular ECM; high collagen density in PDL tissue; high turnover - make and breakdown collagen.

Osteoblasts – located along surface of alveolar bone; build bone.

Osteoclasts – originate from monocytes; located along resorption pits.

Cementoblasts – located along cementum surface; build cementum.

Macrophages – defense cells; phagocytose other cells.

Epithelial rest – remnant of Hertwig's epithelial root sheath (HERS).

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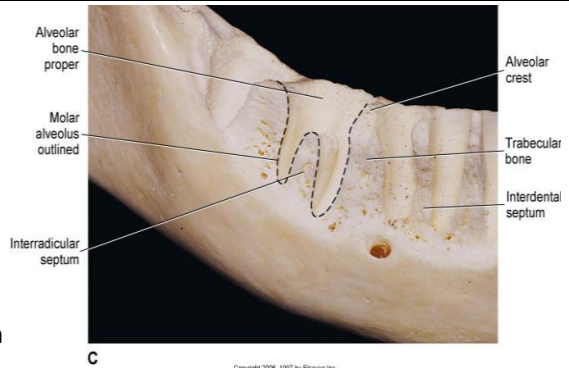
Here is a list and short description of the resident cells of the PDL and a short description of each.

Types of Bone

- 2 sections:
 - **Alveolar Bone:** portion that contains teeth
 - **Basal bone:** portion apical of roots (body)**Both are covered by the periosteum (CT layer)**

Alveolar bone has 2 parts:

- Alveolar Bone Proper (ABP):** Makes up the lining of the tooth socket
- Supporting Alveolar Bone (SAB):** AKA Cortical Plate



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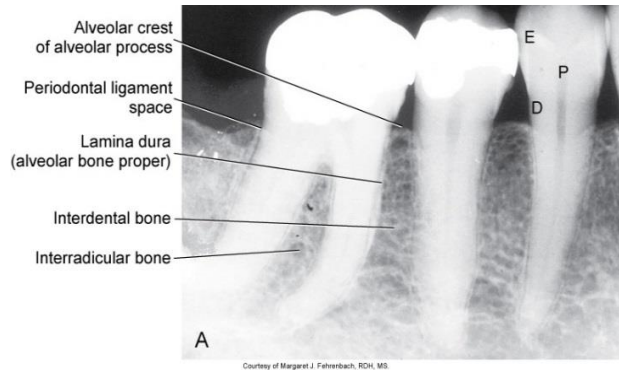
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Courtesy of Dr Javadi

Alveolar Bone Proper

- **Plates of compact bone**
 - Bundle bone (Sharpey's fibers inserted)
 - Lamina Dura (radiopaque on x-rays)
 - Cribriform plate (numerous holes where Volkmann's canals pass from alveolar bone to PDL for nutrients)
- **Alveolar crest:**
 - Coronal border of cervical rim of alveolar bone
 - **In health 1-2 mm apical of CEJ**

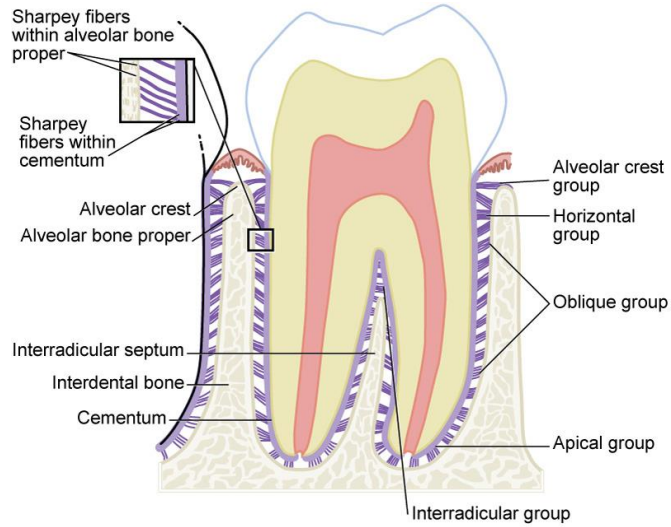
The alveolar process density in a given area determines the dental infection route (during abscess formation) and the efficacy of local anesthetic infiltration.



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Courtesy Dr Javadi

Bundle Bone and Sharpey's Fibers

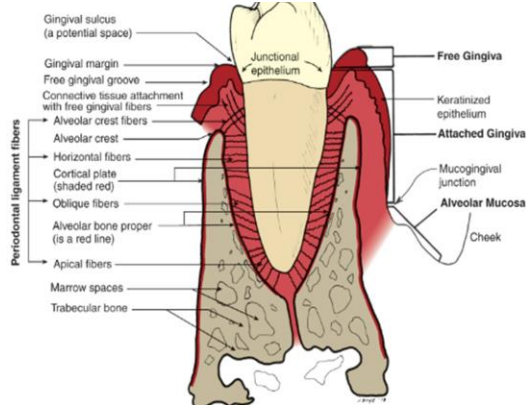


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Courtesy of Dr Javadi

FIGURE 15-42. Periodontium including zones of gingiva and periodontal fiber groups: a mandibular left first premolar is suspended in its alveolus by the five groups of periodontal ligament fibers: *apical, oblique, horizontal, alveolar crest, and free gingival fibers* are visible. A sixth group, called *transeptal fibers*, not visible in this drawing, attaches from the cementum of one tooth to the cementum of the adjacent tooth at a level between the free gingival and alveolar crest fibers. The fibers of the periodontal ligament are much shorter than depicted here, averaging less than $\frac{1}{4}$ of a millimeter.

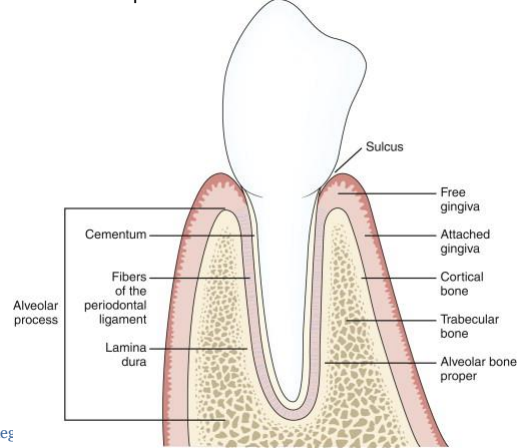


Supporting Alveolar Bone

Consists of both cortical and trabecular bone

-Cortical bone: plate of compact bone on the facial and lingual surfaces

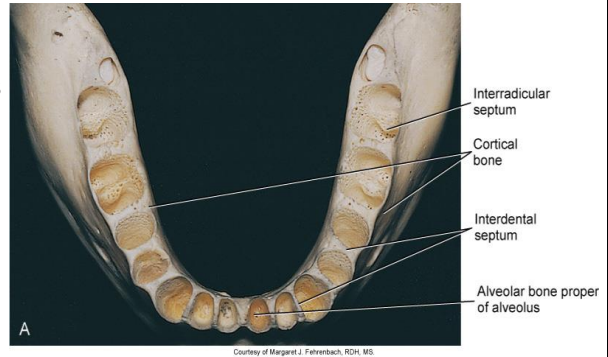
-Trabecular bone: cancellous (spongy) bone between the ABP and plates of cortical



Courtesy of Dr Javadi

Other Bony Landmarks

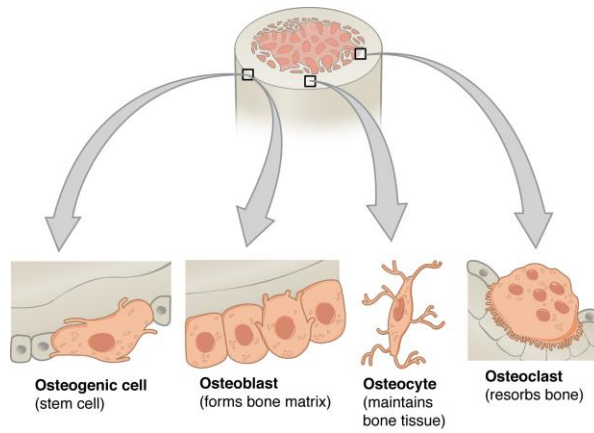
- **Interdental septum or interdental bone:**
 - Alveolar bone between 2 neighboring teeth
 - Consists of compact bone and cancellous bone
- **Interradicular septum or interradicular bone:**
 - Bone between roots of same tooth
 - Consists of alveolar bone proper and trabecular bone



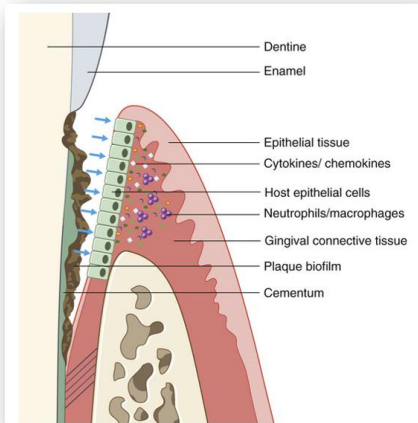
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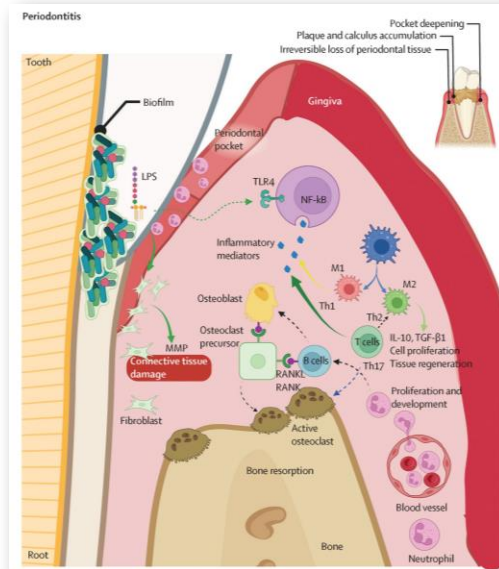
Bone cells



Pathogenesis



- How a disease develops includes origin of the disease, events that take place leading to disease
- Periodontitis : Interaction between Host and Bacteria
- Destruction of tissue=Clinical Attachment Loss, Bone loss, Inflammation, Bleeding



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[Fernando P S Guastaldi](#)

Periodontal disease

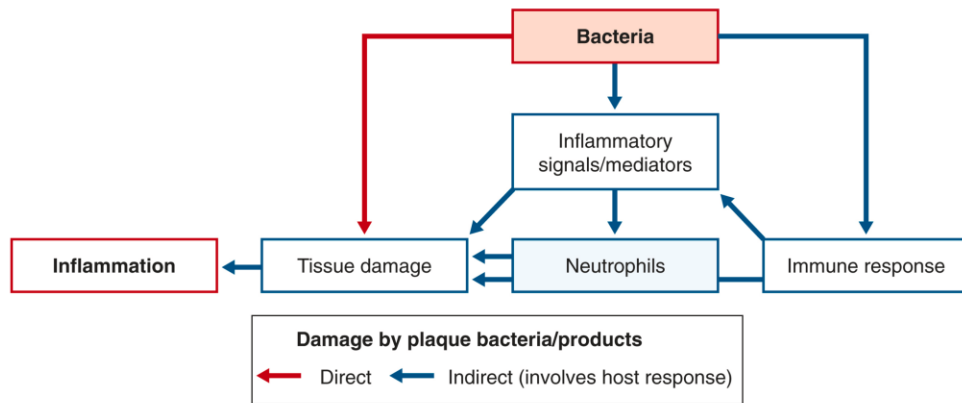
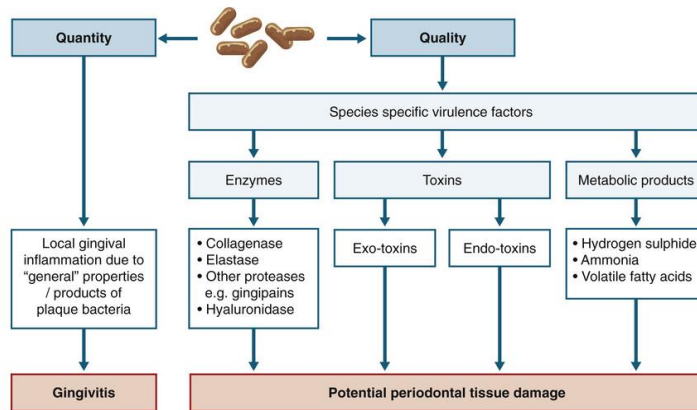


FIGURE 1.2.4 Direct and indirect bacterial effects in periodontitis.

Eaton p. 14

What is happening here? Pathogenesis of tissue damage.

Overview of periodontitis pathogenesis

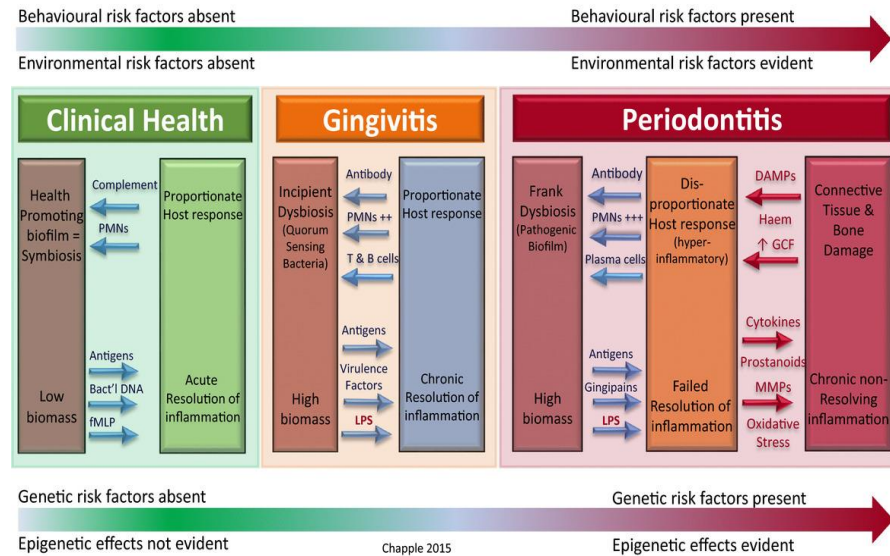


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Fig 1.2.3 Practical Periodontics

Pathogenesis: How a disease develops



Journal of Periodontology, Volume: 89, Issue: S1, Pages: S17-S27, First published: 21 June 2018, DOI: (10.1002/JPER.17-0095)

Contemporary model of host–microbe interactions in the pathogenesis of periodontitis, in which the host response drives an incipient dysbiosis (gingivitis). If the biofilm is not disrupted/removed, frank dysbiosis results and perpetuation.

Bacterial Factors

- Colonization: Biofilm, Dysbiosis
- Quantity: Biomass
- Species-specific virulence factors:

Enzymes

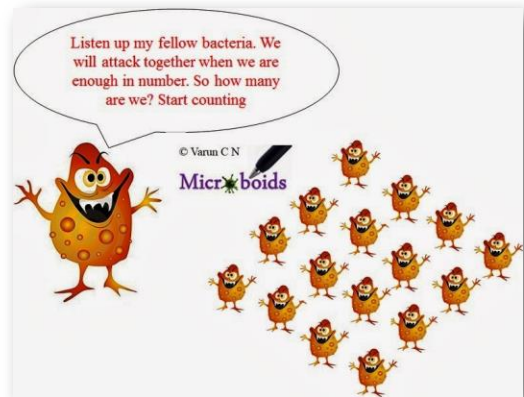
- Collagenase, elastase, proteases

Toxins

- Endotoxin (LPS-lipopolysaccharide), exotoxin

Metabolic products

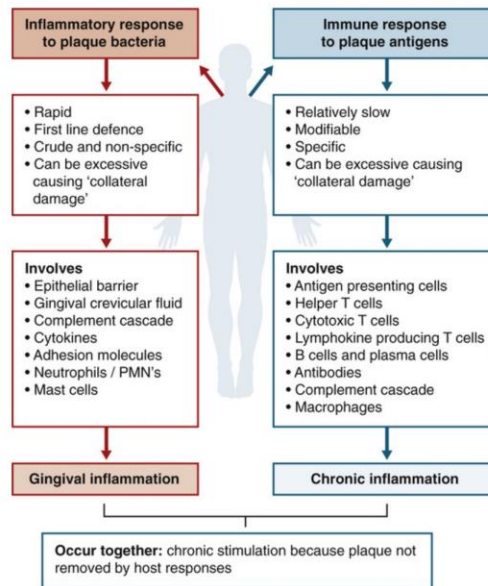
- Hydrogen sulfide, ammonia, volatile fatty acids



Lipopolysaccharide (LPS = bacterial endotoxin) -> infiltration of polymorphonuclear leukocytes -> edema and vasodilation in inflammatory periodontal tissues
LPS modulates the expression levels of Toll-like receptors (TLRs) in the host cells, which regulate the production of proinflammatory molecules could influence on the intensity of inflammation response

Excessive production of cytokines by inappropriate immune response accelerates periodontal tissue destruction

Inflammatory modulators including interleukin (IL)-6, cyclooxygenase (COX) and intercellular adhesion molecule (ICAM) are known to regulate the host response against bacterial infection

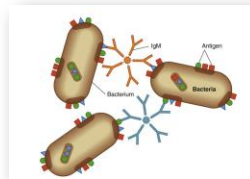
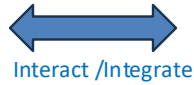


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Host Response

- Inflammatory Response (innate)

- Non-specific
- Epithelial barrier
- Antimicrobial molecules
- Complement cascade
- Cytokines
- Neutrophils
- Mast cells



- Immune Response (adaptive)

- T-cells
- In periodontitis IgG (GCF) & IgA (Saliva) most important
- B-lymphocytes
- Antibodies/Immunoglobulins
 - Plasma cells
- In periodontitis IgG (GCF) & IgA (Saliva) most important

We will discuss Host response more next week

Cytokines: Inflammatory/immune response

- Activation of osteoclasts
- Stimulation inflammatory mediators
- Differentiation of B-cells
- Oxygen radicals, enzymes

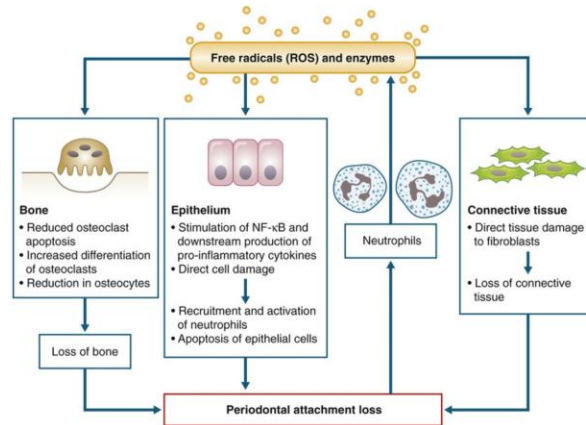


TISSUE DESTRUCTION

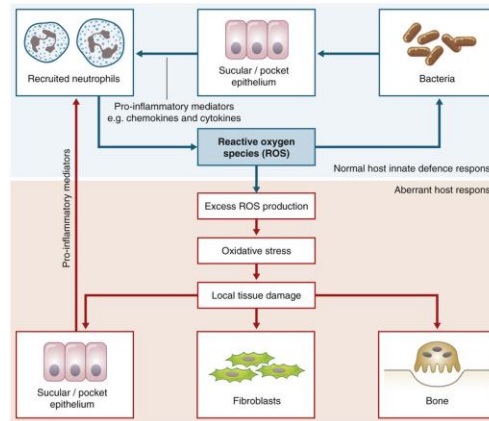
Cytokine	Cell of Origin	Inflammatory/Immune Response
Interleukin 1 (IL-1)	Macrophage (MØ), fibroblast, monocytes upon stimulation by endo/exotoxins, epithelial cells	Activation of osteoclasts ↑ PMNL margination ↑ Prostaglandin PGE2 by fibroblasts ↑ IL-6 synthesis by periodontal fibroblasts ↑ TNF-α production and release
Interleukin 6 (IL-6)	MØ, fibroblasts, epithelial cells	↑ Acute phase protein synthesis by liver ↑ Bone resorption ↑ B-cell differentiation and Ig production ↑ T-cell activation
Interleukin 8 (IL-8)	MØ, endo/epithelial cells, LPS-stimulated fibroblasts, platelets	Potent PMNL chemotaxin
Interleukin 10 (IL-10)		Suppression of cytokines (anti-inflammatory)
Transforming growth factor beta (TGF-β)		Anti-inflammatory (at high concentrations) stimulates collagen synthesis and repair
Tumor necrosis factor alpha (TNF-α)	Activated MØs, monocytes, epithelial cells	↑ MØ production IL-1 ↑ MØ production of PGE-2 ↑ ICAM-1 expression/PMNL margination ↑ PMNL oxygen radical production ↑ PMNL degranulation of enzymes
Prostaglandin E2 (PGE2)	Activated MØs, monocytes, PMNLs, mast cells, epithelial cells	↑ Vascular permeability ↑ Vasodilation ↑ PMNL chemotaxis Stimulates bone resorption
Granulocyte-macrophage colony stimulating factor (GM-CSF)	Epithelial cells, monocytes, endothelial cells, fibroblasts	↑ Release of PMNLs from bone marrow ↑ PMNL chemotaxis ↑ Apoptosis ↑ Degranulation ↑ Oxygen radical formation

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Reactive Oxygen Species (ROS), Enzymes



Oxidative stress and tissue damage



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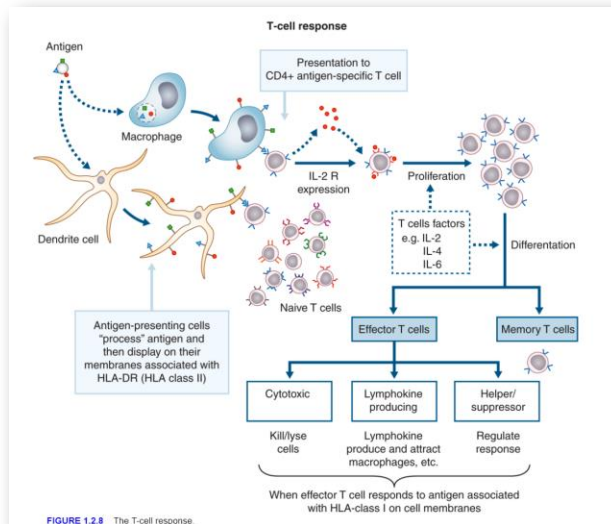
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Inflammation contributes to dysbiosis, inflammatory and immune responses result in tissue damage, and attachment loss. We want to relate this to what we see clinically, inflammatory changes, loss of attachment as indicated by probing depth, bone loss, gingival recession

Immune response

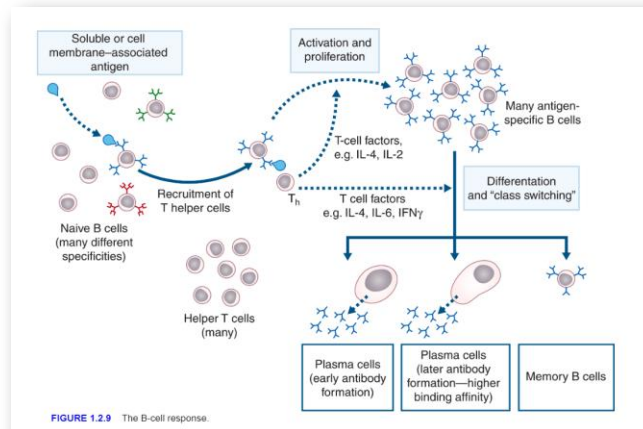
- Antigen presenting cells
- Helper T cells
- Cytotoxic T cells
- Lymphokine-producing T cells
- Antibodies
- Complement cascade
- Macrophages

T-cell response



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B-cell response



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All 5 classes of antibody will be produced, however IgG predominates in periodontitis

Clinical presentation



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Clinical presentation

- Destruction of epithelium, CT and bone
- **Probing depth increases** as clinical attachment is loss (CAL)
- Inflammation, edema, erythema and bleeding (BOP) increases
- Bone loss becomes radiographically apparent
- Tooth mobility, crown to root ratio, furcation invasion, recession may be present

Gingival Sulcus

- Healthy Sulcus:
 - Depth of a clinically normal sulcus is 1-3mm
 - Probing depths are slightly deeper on the mesial and distal surfaces than depths on the buccal and lingual
 - NO bleeding



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The gingival sulcus is the area in between the tooth and the surrounding gingival tissue. A healthy sulcular depth or normal sulcus is one to three millimeters. The probing depths are slightly deeper in the interproximal surface, or on the mesial and distal surfaces of the tooth, rather than on the buccal or lingual surface. A healthy sulcus will not show any bleeding.

Health

1. **Gingival Sulcus.** When healthy, the junctional epithelial cells attach to the enamel of the tooth crown (Figure 12-7).

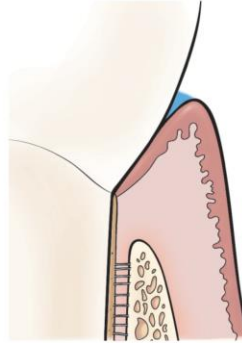


FIGURE 12-7. Gingival Sulcus.

- In health, the junctional epithelium is located slightly apical to—above—the CEJ.
- The junctional epithelium attaches along its entire length to the enamel of the crown.

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Gingivitis

2. **Gingival Pocket.** A **gingival pocket** is a deepening of the gingival sulcus. The increased probing depth seen in a gingival pocket is due to (1) detachment of the coronal portion of the junctional epithelium from the tooth and (2) increased tissue size due to swelling of the tissue (**Figure 12-8**).

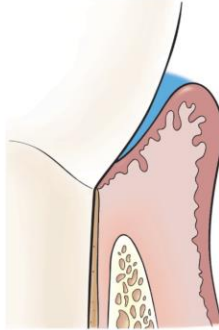


FIGURE 12-8. Gingival Pocket.

- In gingivitis, the coronal-most portion of the junctional epithelium detaches from the tooth, resulting in an increased probing depth.
- Compare the sulcus depth in this illustration with that shown in **Figure 12-7**.
- In some cases of gingivitis, the gingival tissue swells, resulting in an increased probing depth known as a **pseudo-pocket**.

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Periodontal Pocket

3. Periodontal Pocket: A **perforated pocket** forms as the result of the (1) apical migration of the periodontal epithelium and (2) destruction of the periodontal ligament fibers and alveolar bone. **Figure 10.3. Apical migration** is the movement of the apex of the periodontal epithelium. **Figure 10.3. Apical migration** is the movement of the apex of the periodontal epithelium.



- The periodontal pocket, the periodontal epithelium is located on the root below the JE (2).
- The deeper probing depth of periodontal pocket is in the apical region of the gingival sulcus.

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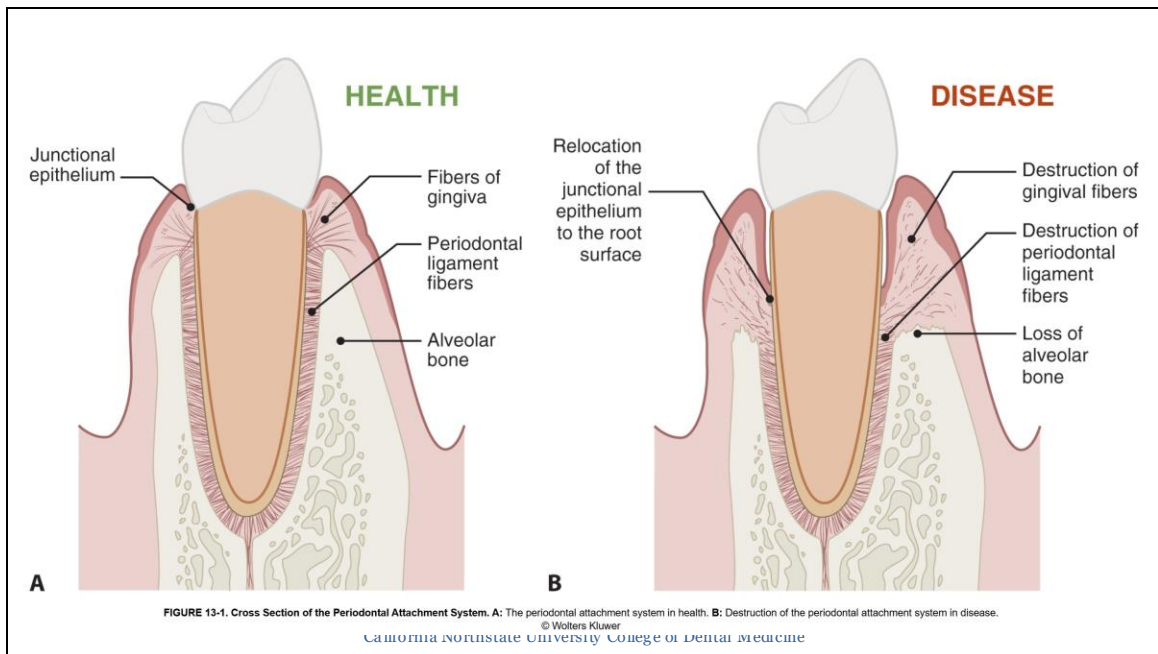
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PERIODONTAL POCKETS

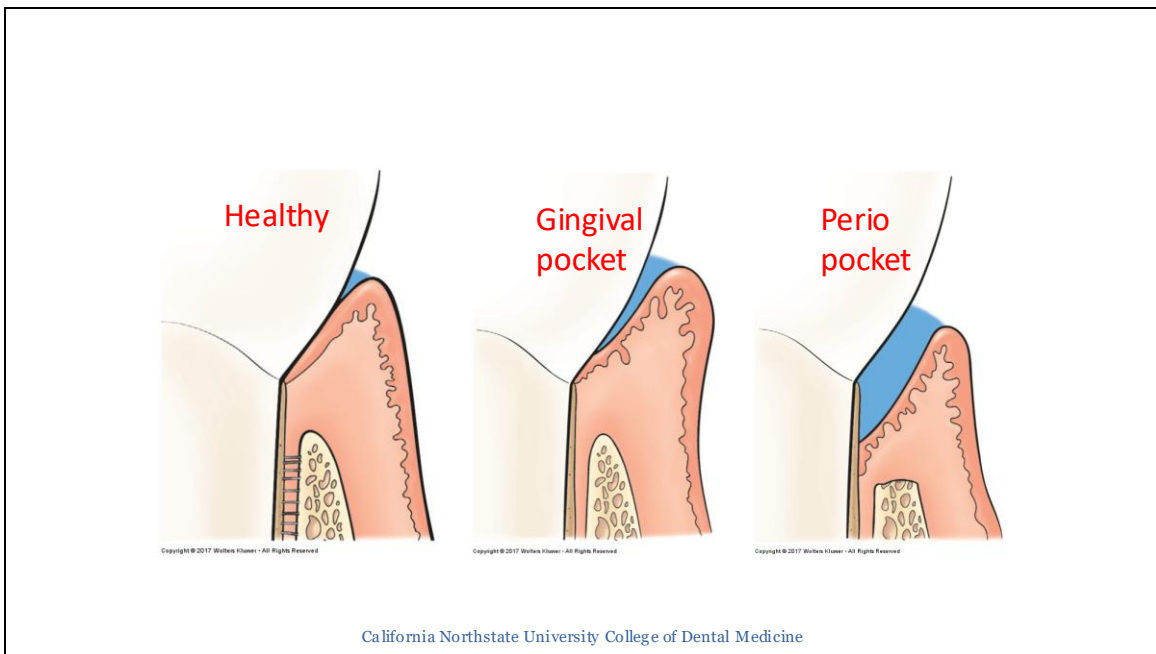
- Periodontal Pocket:
 - Depth of a clinically diseased sulcus is **greater than 3mm**
 - Gingival sulcus that has been deepened by disease
 - Bleeding may or may not be present

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A Periodontal pocket is a diseased gingival sulcus that is deepened due to the apical migration of the junctional epithelium. It usually measures more than 3 mm with a probe and may bleed as a sign of inflammation.



How do you measure a periodontal pocket? You measure the pocket from the base of the pocket to the top of the attached tissue or gingival margin. The cartoon on the left side shows a healthy gingiva with no pocket. The cartoon on the right side shows a diseased pocket where the probe goes deep into the sulcus.



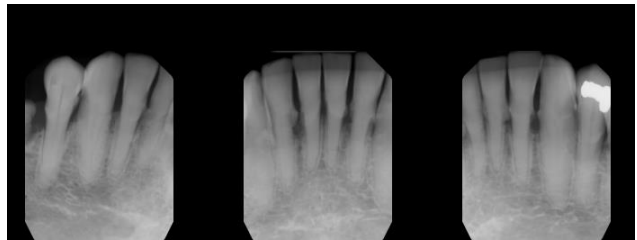
This slide shows the progression of a healthy periodontium (which includes the soft tissue and and bone support) to the initial stage of diseased tissue in a gingival pocket to eventual destruction of bone, seen in the periodontal pocket.

Case #1

- MB is a 73 year old female who presents to the CNU CDM Dental Clinic as a new patient for a limited emergency exam. Her chief complaint is she was eating an apple and now her lower front tooth hurts (she points to #25). Her medical history includes medication for high cholesterol. Her vital signs are BP 138/74.

Assessment of periodontal condition:

- 1. Symptoms of patient**
- 2. Medical and Dental History**
- 3. Visual Assessment**
- 4. Periodontal Examination**
- 5. Radiographic Assessment**
- 6. Assessing local etiologies**

[illegible]

Case #2

- PM is a 62 year-old female that was referred to the CNU CDM Dental Clinic by a local dental hygiene program. She has no chief complaint, and does recall having seen a periodontist about 10 years ago. Her medical history includes estradiol and medroxyprogesterone for hormone replacement therapy. Her vitals are BP 179/84.

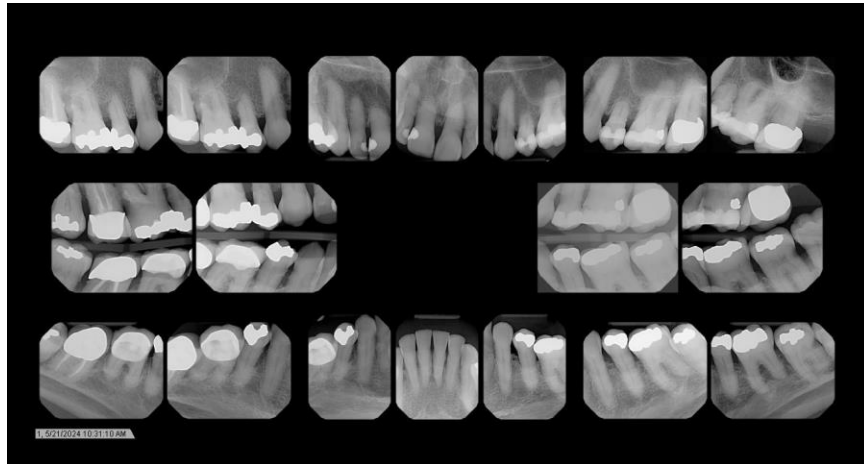
Assessment: Periodontal Examination

1. Probing depths
2. Plaque index
3. Bleeding on probing/exudate
4. Recession
5. **Clinical attachment level**
6. Tooth mobility
7. Furcation involvement
8. Fremitus (a palpable or visible movement of a tooth when subjected to occlusal forces)
9. Occlusion

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The most important part of assessment when it comes to treatment plan, diagnosis is CAL (clinical attachment level/loss)



Radiographic description

Identify:

1. Horizontal bone loss
2. Vertical bone loss
3. Osseous Defects (furcations, craters)
4. Periodontal ligament abnormalities (widened? Peri-apical changes?)
5. Crown to root ratio
6. Root morphology and proximity
7. Calculus deposits
8. Suspected Caries, Status of restorations
9. Tooth Anomalies
10. Impacted teeth
11. Other Lesions/Abnormalities



Learning Objectives

- 1- Describe macroscopic and microscopic anatomy of periodontal tissues (tooth, attachment, bone)
- 2-Apply knowledge to visible anatomy during a clinical exam
- 3-Recognize changes that occur during disease process
- 4-Understand inflammatory response to bacteria
- 5-Understand immune response to bacteria
- 6-Understand the role of host response in pathogenesis

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Review topics covered